

Apache Airflow Assessment 2

Daily Attendance Processor

Objective

The objective of this assessment is to build an Apache Airflow DAG to process daily student attendance data using PythonOperator and file handling concepts.

The workflow performs:

- Create attendance file
- Read attendance records
- Count total students
- Count present students
- Count absent students
- Calculate attendance percentage
- List absent students
- Generate attendance report

DAG Code

```
from airflow import DAG
from airflow.operators.python import PythonOperator
from datetime import datetime
```

```
# Task 1 - Create Attendance File
def create_attendance_file():
```

```
    data = """Aarav,Present
Priya,Present
Rahul,Absent
Sneha,Present
Kiran,Absent
Ananya,Present
Vikram,Present
```

```
Meera,Absent
Farhan,Present
Divya,Present""""
```

```
with open("/tmp/attendance.txt", "w") as file:
    file.write(data)

print("Attendance file created successfully")
```

```
# Task 2 - Read Attendance File
def read_attendance_file():
```

```
    with open("/tmp/attendance.txt", "r") as file:
        data = file.readlines()

    for line in data:
        print(line.strip())
```

```
# Task 3 - Count Total Students
def count_total_students():
```

```
    with open("/tmp/attendance.txt", "r") as file:
        data = file.readlines()

    total_students = len(data)

    print(f"Total Students = {total_students}")
```

```
# Task 4 - Count Present Students
def count_present_students():
```

```
    present_students = 0

    with open("/tmp/attendance.txt", "r") as file:
        data = file.readlines()

    for line in data:

        status = line.strip().split(",")[1]
```

```

    if status == "Present":

        present_students = present_students + 1

    print(f"Present Students = {present_students}")

# Task 5 - Count Absent Students
def count_absent_students():

    absent_students = 0

    with open("/tmp/attendance.txt", "r") as file:
        data = file.readlines()

    for line in data:

        status = line.strip().split(",")[1]

        if status == "Absent":

            absent_students = absent_students + 1

    print(f"Absent Students = {absent_students}")

# Task 6 - Calculate Attendance Percentage
def calculate_attendance_percentage():

    total_students = 10
    present_students = 7

    attendance_percentage = (present_students / total_students) * 100

    print(f"Attendance Percentage = {attendance_percentage}%")

# Bonus Task - List Absent Students
def list_absent_students():

    print("Absent Students List")

```

```
with open("/tmp/attendance.txt", "r") as file:  
    data = file.readlines()
```

```
for line in data:
```

```
    name, status = line.strip().split(",")
```

```
    if status == "Absent":
```

```
        print(name)
```

```
# Task 7 - Generate Attendance Report  
def generate_attendance_report():
```

```
    report = """Daily Attendance Report
```

```
Total Students = 10
```

```
Present Students = 7
```

```
Absent Students = 3
```

```
Attendance Percentage = 70%
```

```
Status = Needs Improvement"""
```

```
with open("/tmp/attendance_report.txt", "w") as file:  
    file.write(report)
```

```
print("Attendance report generated successfully")
```

```
# DAG Definition
```

```
with DAG(
```

```
    dag_id="daily_attendance_processor",
```

```
    start_date=datetime(2025, 1, 1),
```

```
    schedule="@daily",
```

```
    catchup=False,
```

```
    tags=["training"]
```

```
) as dag:
```

```
task1 = PythonOperator(  
    task_id="create_attendance_file",  
  
    python_callable=create_attendance_file  
)
```

```
task2 = PythonOperator(  
    task_id="read_attendance_file",  
  
    python_callable=read_attendance_file  
)
```

```
task3 = PythonOperator(  
    task_id="count_total_students",  
  
    python_callable=count_total_students  
)
```

```
task4 = PythonOperator(  
    task_id="count_present_students",  
  
    python_callable=count_present_students  
)
```

```
task5 = PythonOperator(  
    task_id="count_absent_students",  
  
    python_callable=count_absent_students  
)
```

```
task6 = PythonOperator(  
    task_id="calculate_attendance_percentage",  
  
    python_callable=calculate_attendance_percentage  
)
```

```
task7 = PythonOperator(  
    task_id="list_absent_students",  
  
    python_callable=list_absent_students  
)
```

```
task8 = PythonOperator(  
    task_id="generate_attendance_report",  
  
    python_callable=generate_attendance_report  
)
```

```
task1 >> task2 >> task3 >> task4 >> task5 >> task6 >> task7 >> task8
```